

Emerging Trends

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Overview

- ❖ Sequence to Sequence Models
- ❖ Federated Learning
- ❖ Edge AI
- ❖ Ethics and AI: Responsible AI

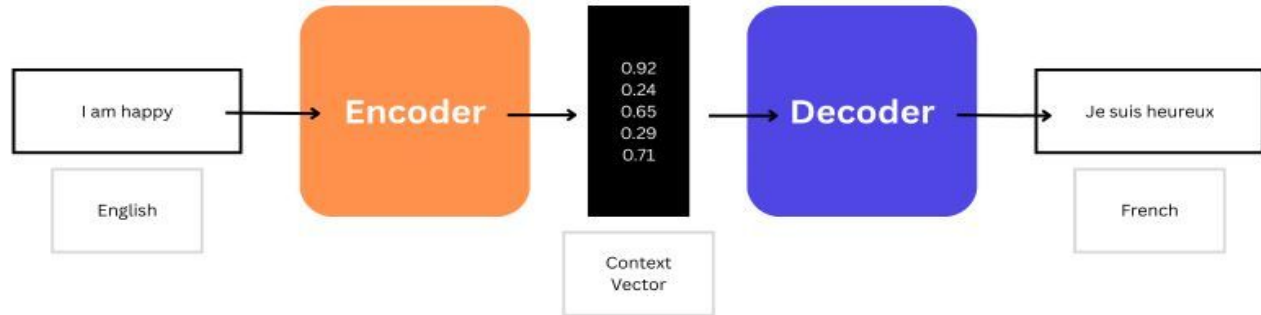
Sequence to Sequence Models

- ❖ A Sequence-to-Sequence (Seq2Seq) model is a deep learning architecture designed to map an input sequence of arbitrary length to an output sequence of a different arbitrary length.
- ❖ Why Encoder and Decoder?
 - Traditional neural networks (like feedforward or basic convolutional nets) map a fixed-size input vector directly to a fixed-size output vector.
 - However, real-world sequential data (language, audio, time-series) violates this rigid structure in two major ways:
 - Variable and Unmatched Lengths
 - Non-Monotonic Alignment
 - The Encoder-Decoder architecture solves this by decoupling the processing of the input sequence from the generation of the output sequence.

Sequence to Sequence Models

Encoder: The Encoder's responsibility is to convert a sequence of discrete tokens into continuous vector representations that capture semantic, syntactic, and contextual relationships.

Decoder: The Decoder's responsibility is to generate the output sequence token-by-token autoregressively until a termination token (<EOS>) is generated.



Application of Sequence to Sequence Model

- ❖ **Machine Translation** — e.g., Google Translate (English → Nepali), the original and most famous Seq2Seq use case.
- ❖ **Text Summarization** — condensing long documents into short summaries (abstractive summarization).
- ❖ **Chatbots and Conversational AI** — mapping a user query sequence to a response sequence.
- ❖ **Speech Recognition** — mapping an audio waveform/feature sequence to a text transcript.
- ❖ **Speech Synthesis (Text-to-Speech)** — mapping text sequences to audio sequences.
- ❖ **Question Answering** — generating a free-form answer sequence given a question and context.
- ❖ **Code Generation** — translating natural language instructions into source code.
- ❖ **Image Captioning** (a variant where the encoder is a CNN and the decoder is an RNN/Transformer) — mapping an image to a descriptive sentence.

Sequence to Sequence

- ❖ RNNs / LSTMs (2014-2015)
- ❖ Attention-Seq (2015-2017)
- ❖ Transformers (2017-Present)
- ❖ State Space Models (Mamba / 2021+)

Federated Learning (FL)

- ❖ Federated Learning (FL) is a distributed machine learning paradigm in which a model is trained across many decentralized devices or servers holding local data, without that raw data ever leaving its source device.
- ❖ How it typically works
 - A **central server** initializes a **global model** and sends a copy to a selected subset of participating devices/clients (e.g., smartphones, hospitals, banks).
 - **Each client trains the model locally** for a **few steps/epochs** using only its own **local data**.
 - **Each client sends back** only the **updated model parameters** (weights/gradients) — never the raw data — to the central server.
 - **The server aggregates** these updates (commonly by averaging them, weighted by the amount of local data each client used) to produce an **improved global model**.
 - This cycle repeats over many communication rounds until the global model converges.

Federated Learning (FL): Benefits

- ❖ **Data privacy and confidentiality** — sensitive raw data (medical records, personal messages, financial transactions) never leaves the local device, addressing a core weakness of centralized ML.
- ❖ **Regulatory compliance** — helps satisfy data protection laws such as GDPR and HIPAA, which restrict movement/centralization of personal data.
- ❖ **Reduced bandwidth/communication cost** — only model updates (much smaller than raw datasets) are transmitted, not entire data dumps.
- ❖ **Access to otherwise unusable data** — enables training on data that legally or practically could never be centralized (e.g., cross-hospital medical data, cross-bank fraud data).
- ❖ **Personalization** — the global model can be further fine-tuned locally, allowing devices to have both shared general intelligence and device-specific personalization.
- ❖ **Real-time/on-device learning** — models can improve continuously from live, local user behavior.

Federated Learning (FL): Application

- ❖ **Mobile keyboard prediction** — Google's Gboard uses FL to improve next-word prediction using data typed on millions of phones, without Google ever seeing what users typed.
- ❖ **Healthcare** — training diagnostic models (e.g., tumor detection) collaboratively across multiple hospitals without sharing patient records.
- ❖ **Finance** — collaborative fraud detection models across banks without exposing individual customer transaction data.
- ❖ **IoT and smart devices** — learning usage patterns across a fleet of devices (smart speakers, wearables) while preserving user privacy.
- ❖ **Autonomous vehicles** — fleets of cars collaboratively improving perception/driving models using locally collected driving data.

Edge AI

- ❖ Edge AI refers to running AI/ML models directly on local ("edge") devices — such as smartphones, IoT sensors, cameras, drones, or embedded microcontrollers — as opposed to sending data to a centralized cloud server for processing and waiting for a response.
- ❖ Computation happens at or near the point where data is generated, i.e., "at the edge" of the network, rather than in a distant data center.
- ❖ **Key factor:**
 - Model compression techniques that make large models small and fast enough to run on constrained hardware:
 - **Quantization** — reducing numerical precision of weights (e.g., 32-bit floats → 8-bit integers) to shrink model size and speed up inference.
 - **Pruning** — removing redundant or low-importance neurons/weights from a trained network.
 - **Knowledge Distillation** — training a small "student" model to mimic the behavior of a large "teacher" model.
 - Specialized edge hardware: NPUs (Neural Processing Units), TPUs, GPUs in mobile SoCs, and dedicated AI accelerator chips (e.g., Google Coral, NVIDIA Jetson).
 - Lightweight frameworks: TensorFlow Lite, ONNX Runtime, PyTorch Mobile — optimized runtimes for deploying models on edge hardware.

Edge AI: Benefits

- ❖ **Low latency** — decisions are made in milliseconds locally, critical for applications like autonomous driving or industrial safety systems where waiting for a round-trip to the cloud isn't acceptable.
- ❖ **Reduced bandwidth and cost** — only relevant, processed insights need to be sent to the cloud (or nothing at all), rather than continuous raw sensor/video streams.
- ❖ **Improved privacy and security** — raw data can be processed and discarded locally, never transmitted over networks, reducing exposure to interception or breach.
- ❖ **Offline / intermittent-connectivity operation** — devices continue functioning even without stable internet, important for remote or mobile deployments.
- ❖ **Scalability** — distributing computation across many edge devices reduces the load and cost on centralized cloud infrastructure.
- ❖ **Energy efficiency at scale** — avoids the energy cost of constantly transmitting large volumes of raw data over networks.

Edge AI: Application

- ❖ **Smartphones** — on-device face unlock, voice assistants (wake-word detection), real-time photo enhancement.
- ❖ **Autonomous vehicles** — real-time object detection, lane detection, and obstacle avoidance processed on-board.
- ❖ **Industrial IoT / manufacturing** — real-time defect detection on production lines, predictive maintenance using vibration/sound sensors.
- ❖ **Smart surveillance cameras** — on-device person/object detection without streaming continuous video to the cloud.
- ❖ **Healthcare wearables** — real-time heart-rate anomaly or fall detection processed directly on the wearable device.
- ❖ **Agriculture** — drone- or sensor-based crop health/pest detection in remote fields with poor connectivity (relevant to Nepal's agricultural/rural context).
- ❖ **Retail** — smart shelves and checkout-free stores performing real-time inventory/vision analysis in-store.

Responsible AI

- ❖ Responsible AI is a governance framework and ethical approach to designing, building, and deploying artificial intelligence systems that are safe, trustworthy, and beneficial to society.
- ❖ It moves beyond just asking "what can we build?" to "what should we build?" ensuring that AI systems respect human rights and democratic values.
- ❖ The core pillars of this approach typically include
 - Fairness (ensuring the AI does not discriminate)
 - Accountability (establishing who is responsible for the AI's actions)
 - Transparency (making the system's operations understandable)
 - Privacy (protecting user data),
 - Safety (ensuring the system doesn't cause physical or digital harm).

Framework for Developing and Deploying AI

- ❖ A robust framework for developing and deploying AI involves a structured lifecycle that embeds ethical checks at every stage, from concept to retirement.
- ❖ It begins with Governance, where a diverse committee establishes policies and risk thresholds before any code is written.
- ❖ During the Design and Data Collection phase, teams must document the provenance of their data (using tools like "Datasheets for Datasets") to ensure consent and quality.
- ❖ In the Development and Training phase, models are subjected to "Red Teaming"—where ethical hackers try to break the model or force it to generate harmful content—to identify vulnerabilities.
- ❖ Finally, in the Deployment and Monitoring phase, the system is not just released but continuously watched for "model drift," where the AI's performance degrades or becomes biased over time as real-world data changes.

Concepts of Bias and Fairness in AI

- ❖ **Bias** in AI refers to systematic errors that create unfair outcomes, often privileging one group of users over others.
 - This can stem from **Historical Bias** (where the training data reflects past prejudices)
 - **Sampling Bias** (where the data doesn't represent the real world, like a facial recognition system trained mostly on light-skinned faces failing to recognize dark-skinned faces).
- ❖ **Fairness** is the proactive correction of these biases. It is complex because there are different mathematical definitions of "fair."
 - Group Fairness ensures that the AI's approval rates are equal across different demographic groups (e.g., men and women get loans at the same rate)
 - Individual Fairness ensures that similar individuals are treated similarly regardless of their group.

Explainable Artificial Intelligence(XAI)

- ❖ Explainable artificial intelligence (XAI) refers to a collection of procedures and techniques that enable machine learning algorithms to produce output and results that are understandable and reliable for human users.
- ❖ Explainable AI is a key component of the fairness, accountability, and transparency (FAT) machine learning paradigm and is frequently discussed in connection with deep learning.
- ❖ Benefits of explainable AI
 - Improved decision-making
 - Increased trust and acceptance
 - Reduced risks and liabilities
- ❖ Explainable AI principles
 - Transparency
 - Interpretability
 - Accountability

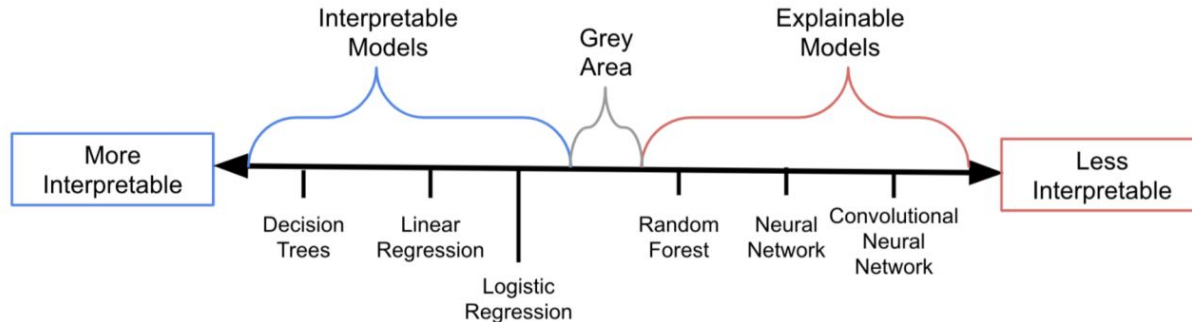
Explainable Artificial Intelligence(XAI)

❖ Explainable AI approaches

- Feature importance
- Attribution
- Visualization

❖ Explainable AI (XAI) Techniques

- LIME (Local Interpretable Model-agnostic Explanations)
- SHAP (SHapley Additive exPlanations)
- ELI5 (Explain Like I'm 5)



Ethical Dilemmas in AI

❖ Job Displacement and Automation

- The most immediate ethical dilemma is the economic disruption caused by automation.
- Unlike previous industrial revolutions that replaced physical labor, AI is unique because it replaces cognitive labor.
- This creates a risk of "technological unemployment," where machines outperform humans in tasks ranging from truck driving to coding and medical diagnostics.
- For example, a report by McKinsey estimates that up to 800 million jobs could be displaced by 2030.

❖ Misuse of AI: Deep Fakes and Disinformation

- The democratization of powerful AI tools has led to significant misuse, particularly through Deepfakes—synthetic media that swaps faces or voices with frightening realism.
- This poses a threat to personal reputation, political stability, and truth itself.
- In the political sphere, we have already seen AI-generated "robocalls" mimicking political leaders (like President Biden) to discourage voting.
- In the financial sector, deepfake audio of CEOs has been used to trick employees into transferring millions of dollars to fraudsters.
- This creates a "Liar's Dividend," a societal state where because anything could be fake, people stop believing real evidence, eroding trust in journalism and judicial systems.

Societal Impact and Governance

❖ Societal Impact: Bias and Surveillance

- On a societal level, AI can amplify existing prejudices if not carefully governed.
- Algorithms used in predictive policing (like the controversial PredPol) have been found to target minority neighborhoods disproportionately because they are trained on historical arrest data, which itself contains human bias.
- Similarly, in healthcare, algorithms used to allocate care have historically underestimated the needs of Black patients because the model used "healthcare costs" as a proxy for "health needs," ignoring the fact that minority groups often have less access to spending.
- Furthermore, the rise of facial recognition creates a Surveillance State dilemma, where anonymity in public spaces is lost.
- Governments can theoretically track a protestor's movement across an entire city instantly, chilling free speech and the right to assembly.

Regulatory Frameworks and Governance

- ❖ The EU AI Act (The "Risk-Based" Model):
 - This is the world's first comprehensive AI law.
 - It categorizes AI into risk levels.
 - Unacceptable Risk (like social scoring systems or real-time biometric surveillance in public) is banned entirely.
 - High Risk (like AI in surgery, hiring, or border control) is heavily regulated with strict transparency and human oversight requirements.
- ❖ US Executive Order on AI (The "Innovation & Safety" Model):
 - The US approach focuses more on industry standards and national security.
 - It mandates that companies developing "dual-use" foundation models (like GPT-4) must share safety test results ("Red Teaming") with the government before public release.
 - It emphasizes maintaining a competitive edge while ensuring safety.
- ❖ China's AI Regulations (The "State Control" Model):
 - China has rolled out specific rules for Generative AI, requiring that AI outputs must align with core socialist values and that providers are responsible for the content their AI generates.
 - It heavily emphasizes control over information flow and political stability.

AI Policies

- ❖ Trust and Transparency:
 - All the websites emphasize the importance of trust and transparency in AI systems.
 - This involves explaining how technology is used in decision-making, ensuring decisions are auditable and open for inspection, and disclosing how data is used and stored.
 - For instance, Deloitte's Trustworthy AI™ framework highlights the need for AI to be transparent and explainable, with a focus on fair and impartial use, robustness and reliability, and respecting user privacy.
- ❖ Ethical Principles and Guidelines:
 - Each organization advocates for AI to adhere to ethical principles and guidelines.
 - Common themes include the avoidance of harm, the promotion of fairness and justice, and the preservation of autonomy.
 - Deloitte, for example, discusses core principles like impact, justice, and autonomy, echoing major schools of ethical philosophy.
- ❖ Responsibility and Accountability:
 - There's a strong focus on establishing who is responsible and accountable for the outputs of AI systems.
 - This includes identifying the roles of various stakeholders like developers, testers, and management in ensuring ethical use of AI.
 - The Deloitte AI Institute's Trustworthy AI framework, for example, includes a dimension specifically for responsibility and accountability.

AI Policies

- ❖ Addressing Bias and Fairness:
 - The need to actively identify and mitigate biases in algorithms and data is a recurring topic.
 - This is crucial to prevent AI systems from perpetuating or amplifying societal biases.
- ❖ Safety and Security:
 - Ensuring the safety and security of AI systems is another common element.
 - This includes protection from cybersecurity risks and ensuring that AI does not cause physical or digital harm.
- ❖ Adaptability and Continuous Learning:
 - There's an understanding that AI ethics is a rapidly evolving field.
 - Hence, there's a need for continuous learning and adaptation of policies and frameworks as new ethical challenges and technological advancements emerge.
- ❖ Legal and Regulatory Compliance:
 - Compliance with existing laws and regulations is a foundational aspect.
 - Organizations stress the importance of aligning AI practices with legal standards and preparing for future regulations in the field of AI ethics.

Privacy & Security in AI

❖ Privacy Risks Associated with AI

- The massive data requirements of AI systems create significant privacy vulnerabilities that go beyond traditional data breaches.
- The core risk is that AI models can memorize sensitive information from their training data and inadvertently reveal it.
- Data Privacy in AI is not just about encryption but about "unlearning" sensitive details while keeping useful patterns.
- One major threat is the **Inference Attack**, where an attacker queries a model to deduce private information.
- A specific type is the **Membership Inference Attack (MIA)**, where a hacker determines if a specific person's data (like a patient's medical record) was used to train a model, effectively revealing their diagnosis.
- Another is **Model Inversion**, where an attacker reconstructs the original training data—such as recreating a person's face from a facial recognition system—just by analyzing the system's confidence scores.

Privacy & Security in AI

❖ Data Anonymization Techniques

- To combat these risks, developers use advanced anonymization techniques.
- Traditional methods like removing names (masking) are often insufficient because AI can cross-reference data to re-identify people.
- Differential Privacy is the gold standard today; it adds mathematical "noise" to the dataset so that the AI learns general trends (e.g., "smoking causes cancer") without learning any specific individual's data (e.g., "John Doe smokes").
- Another technique is **Federated Learning**, where the AI model is sent to the user's device (like a phone) to learn from data locally. Only the lessons learned (mathematical updates) are sent back to the central server, not the raw photos or messages, ensuring personal data never leaves the user's hands.

AI Security Threats

- ❖ AI systems face unique security threats that target the "brain" of the model rather than just the server it runs on.
- ❖ **Adversarial Attacks** (or Evasion Attacks) involve subtly modifying input data to trick the AI.
- ❖ **For example**, hackers can place a specifically patterned sticker on a Stop sign; to a human, it's just a sticker, but to a self-driving car's computer vision, it might look like a "Speed Limit 45" sign, causing a crash.
- ❖ **Data Poisoning** is an attack during the training phase where hackers inject malicious data into the training set.
- ❖ A famous example is Microsoft's "Tay" chatbot, which Twitter users "poisoned" by feeding it racist and inflammatory tweets, causing the bot to become offensive within 24 hours.

Defense Strategies

- ❖ Defending AI requires a "Defense in Depth" approach.
- ❖ **Adversarial Training** is a brute-force method where developers intentionally attack their own models with adversarial examples during training, teaching the AI to recognize and ignore them.
- ❖ **Input Sanitization** involves cleaning data before it reaches the model, such as blurring images slightly to destroy the precise pixel patterns that adversarial attacks rely on.
- ❖ For poisoning, **Outlier Detection** algorithms scan training data for suspicious inputs (like a massive spike in similar requests) and discard them before they can corrupt the model.
- ❖ Additionally, **Model Monitoring** ensures that if a model starts behaving erratically (drifting) in the real world, it is automatically shut down for inspection.